

APPLICATION SECTION

QUANTIFICATION OF A MALFUNCTION IN A WEAPON SYSTEM

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Summary: In this paper a particular malfunction in a weapon system is discussed. A model is proposed to study and analyse this malfunction in order to aid in the decision making process. Some applications of this model to practical situations are given.

1. INTRODUCTION

The gunnery problem at sea can be described as the problem of getting a shell to hit a moving target many kilometres away, from a ship that is also moving through the water rolling and pitching.

A modern gunnery system is a very complicated electronic-mechanical system that tries to solve the above problem (see, for example, Frieden 1985). We will give a brief description of such a system in Section 2 and show that the variation in fall of shot is made up of a series of smaller component sources of variation which in turn can broadly be classified into three main sources of variation, namely those due to tracking, prediction and ballistics.

This paper will be concerned with ascertaining the effect of a certain

mechanical malfunction that occurred in a particular system. This malfunction introduced a systematic error in the tracking component and showed up as distinct peaks on pen records measuring the errors of the gun pointing position relative to the aiming instruction received from the fire control computer. These peaks occurred in elevation only, when a ship changes the direction in which it is rolling, leading alternatively to positive and negative peaks. In the subsequent discussion we will refer to these as EORP's (end-of-roll peaks). A typical pen record showing distinct EORP's is depicted in Figure 1.1 below.

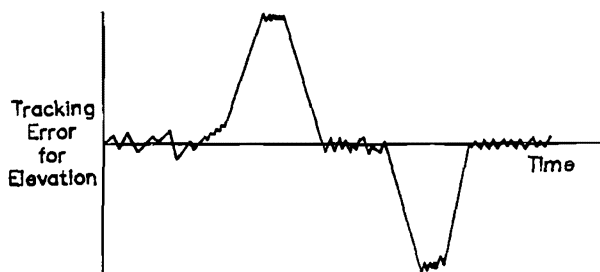


Figure 1.1: Pen records showing EORP's

Clearly the occurrence of such peaks could have a serious effect on system effectiveness. However, modifying the system to eliminate (or reduce) EORP's is quite costly and it was not clear to our client whether this cost was justified. We were tasked to quantify the effect of the EORP's in order to supply them with better grounds for making a decision. In this paper we report on the effect of the EORP's on the hit probabilities for certain realistic situations.

In the next section we give a brief system description and in Section 3 we discuss the model and its analysis. Two examples are discussed in Section 4 and some comments are made in the last section.

2. SYSTEM DESCRIPTION

In this section we give a brief description of a typical gunnery system in very general terms.

A gunnery system can be divided into three major sub-systems, namely a tracking radar, a fire control computer and a gun. We discuss the flow of information through the system with the aid of Figure 2.1 below. For the sake of simplicity we do not indicate the feedback of information between sub-systems.

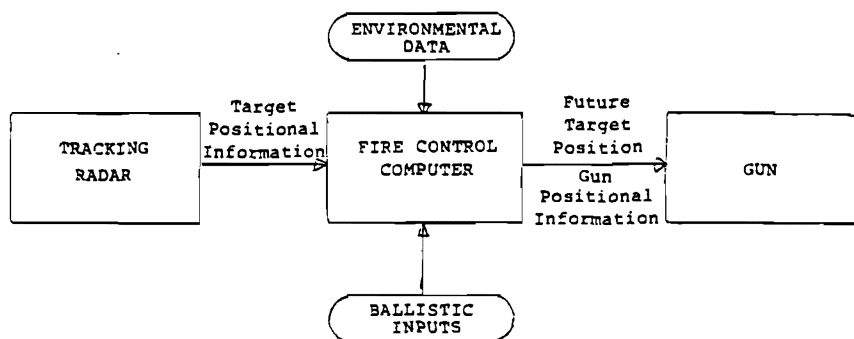


Figure 2.1 : Flow of information through a gunnery system

After a target has been detected (for example by a search radar) and locked onto by the tracking radar, target position information is continuously fed to the fire control computer. At the same time the fire control computer is fed with a variety of other information, for example data on wind strength and direction, ship motion, temperature, pressure and ballistics. This information is processed by the fire control computer to estimate a future target position and converted into a gun pointing position. It is well known that the distribution of fall of shot for a gun firing at

a fixed angle at a stationary surface target can be assumed to be Gaussian with mean the midpoint of the target and variances depending on range of fire and the ballistics of the system. It is clear from the above that for a nonstationary target additional variation is introduced by a series of constituent sources of variation which can be broadly classified as that due to tracking and that due to prediction. (Note that the latter can be considered to be negligible for surface firing). In this case it is quite reasonable to assume for a properly functioning system, that the distribution of the fall of shot is still Gaussian with the same mean but inflated variances. However, in our case a sizeable systematic error occurs in tracking due to EORP's for elevation. These errors are usually measured in minutes of arc and can be transformed into a metric range error using Range Tables (a ballistic trajectory model) for surface firings. The height of peaks and time intervals between peaks to a large extent depend on the severity and frequency of ship roll and therefore on seastate. We now discuss a model which we used to assess the effect on EORP's on the hit probabilities of the gunnery system.

3. MODEL AND ANALYSIS

Consider a single burst of k shots fired at a surface target. Let $T_1, T_2, \dots, T_k$ be the firing times of the k shots. Denote the range errors in fall of shot in the case of no peaks by $Z(T_1), Z(T_2), \dots, Z(T_k)$. We assume that the $\{Z(T_j)\}$ (for fixed $T_1, T_2, \dots, T_k$) each have a Gaussian distribution with mean zero (that is no system bias) and constant variance σ^2 . From the available data records the peaks can be very well approximated by a triangular function, that is the size of the peak grows linearly at a certain rate until it reaches its maximum and then decreases at the same rate until it reaches zero. Let $c(t)$ denote a triangular function with base d and height h :

$$(1) \quad \begin{aligned} c(t) &= 2hd^{-1}(t-r+d) & r-d < t \leq r-\frac{1}{2}d \\ &= 2hd^{-1}(r-t) & r-\frac{1}{2}d < t \leq r \end{aligned}$$

Here we used the following notation

r : peak repetition times (in seconds)

d : peak duration (in seconds)

h : peak height (in metres)

The function $c(t)$ is shown schematically in Figure 3.1.

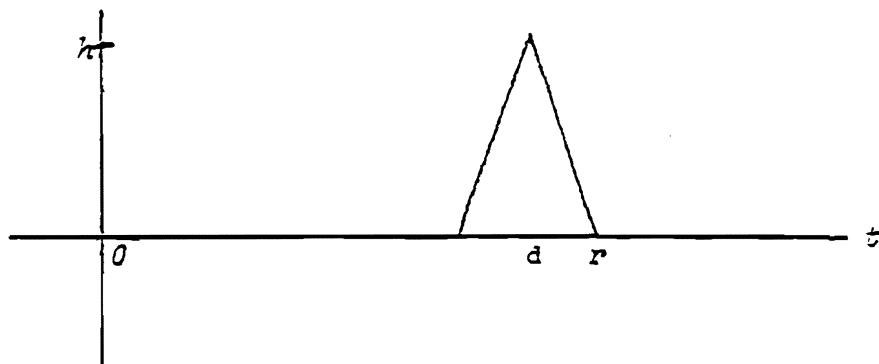


Figure 3.1 : The function $c(t)$

The peaks are alternatively positive and negative and thus approximated by $c(t)$ and $-c(t)$, respectively.

To simplify the notation and exposition, we assume that we have rescaled the $\{T_j\}$ so that they all lie in the interval $(0, r)$. Let ϵ denote the (fixed) interfering time, that is the time between shots. Then the time of the first shot T_1 is taken in $(0, r)$ and the firing times of the subsequent shots are given by

$$(2) \quad T_j = (T_{j-1} + \epsilon) - r[T_{j-1} + \epsilon]/r \quad j = 2, 3 \dots, k$$

with $[x]$ denoting the largest integer function. This follows easily by noting that if a firing time exceeds r it is rolled back until it lies in $(0, r)$. Note that we assume a constant peak repetition time. This is a reasonable assumption to make in many practical situations.

Let $X(T_j)$ be the point of impact (in range) of the j -th shot. Then

$$X_j \equiv X(T_j) = Z(T_j) \pm c(T_j)I(r-d < T_j \leq r), \quad j = 1, 2, \dots, k,$$

with $I(\cdot)$ the indicator function and the $\pm$ denoting respectively a positive or negative peak.

A very reasonable assumption to make is that T_1 is uniformly distributed over $(0, r)$ (and stochastically independent of $Z(\cdot)$) and we will assume this throughout the paper. Given T_1 , the times $T_2, T_3, \dots, T_k$ are fixed since we have a fixed interfiring time ϵ . As a measure of the effect of EORP's we use the expected number of hits in k shots. We now obtain an expression for this.

Suppose that the effective target position for range is $[-a, b]$. In this we include the so called danger space, that is the effective length of the target with the influence of its height on this included.

Define

$$\begin{aligned} Y_j &= 1 \text{ if } X_j \in [-a, b] \\ &= 0 \text{ otherwise} \end{aligned}$$

and let

$$S_k = \sum_{i=1}^k Y_i$$

be the number of hits in k shots. The following result gives an expression for ES_k .

Theorem. Assume the first peak is positive or negative with probability $\frac{1}{2}$ each.

Then

$$\begin{aligned} (3) \quad ES_k &= \sum_{j=1}^k r^{-1} \{ (\Phi(b/\sigma) - \Phi(-a/\sigma)) L(\{t : T_1 = t \text{ and } T_j \notin (r-d, r)\}) \\ &\quad + \frac{1}{2} \int_0^r I(r-d < T_j \leq r) (\Phi((b-c_{jt})/\sigma) - \Phi((-a-c_{jt})/\sigma) \\ &\quad + \Phi((b+c_{jt})/\sigma) - \Phi((-a+c_{jt})/\sigma)) dt \} \end{aligned}$$

with $c_{jt} \equiv c(T_j)$ taking T_1 fixed at t , Φ the standard Gaussian distribution function and $L(A)$ the Lebesgue measure of the set A .

Proof. We have

$$(4) \quad \begin{aligned} ES_k &= \sum_{j=1}^k EY_j = \sum_{j=1}^k P(-a \leq X_j \leq b) \\ &= \sum_{j=1}^k r^{-1} \int_0^r P(-a \leq X_j \leq b | T_1 = t) dt. \end{aligned}$$

Note that $T_j, j = 2, 3, \dots, k$ are fixed functions of T_1 , so for T_1 fixed at t , they are also fixed and $\pm c(T_j)$ is implicitly a (nonrandom) function of t .

Now, with $Z_j \equiv Z(T_j)$ we have

$$\begin{aligned} &P(-a \leq X_j \leq b | T_1 = t) \\ &= \frac{1}{2} \{ P(-a \leq Z_j + c_{jt} I(r-d < T_j < r) \leq b | T_1 = t) \\ &\quad + P(-a \leq Z_j - c_{jt} I(r-d < T_j < r) \leq b | T_1 = t) \} \\ &= \frac{1}{2} \{ \Phi((b - c_{jt} I_j(t))/\sigma) - \Phi((-a - c_{jt} I_j(t))/\sigma) \\ &\quad + \Phi((b + c_{jt} I_j(t))/\sigma) - \Phi((-a + c_{jt} I_j(t))/\sigma) \} \quad (\text{with } T_1 = t) \\ &= (\Phi(b/\sigma) - \Phi(-a/\sigma)) I(T_j \notin (r-d, r)) \\ &\quad + \frac{1}{2} \{ \Phi((b - c_{jt})/\sigma) - \Phi((-a - c_{jt})/\sigma) \\ &\quad + \Phi((b + c_{jt})/\sigma) - \Phi((-a + c_{jt})/\sigma) \} I_j(t) \end{aligned}$$

where $I_j(t) = I(r-d < T_j < r)$ (with $T_1 = t$).

Substituting this in (4) and integrating we obtain the desired result.

Remark. Using (2) and remembering that c_{jt} is a function of t , we see that the integral in the RHS of (3) becomes a sum of integrals of the form

$$(5) \quad \int_{\alpha}^{\beta} \{ \Phi((b - c(t))/\sigma) - \Phi((-a - c(t))/\sigma) \\ + \Phi((b + c(t))/\sigma) - \Phi((-a + c(t))/\sigma) \} dt$$

These integrals can be evaluated using the following result.

Lemma. For $\alpha \leq \beta, u \neq 0, v \in R$

$$\begin{aligned}
 (6) \quad \int_{\alpha}^{\beta} \{\Phi(ux+v)dx &= (\beta + vu^{-1})\Phi(u\beta + v) \\
 &- (\alpha + vu^{-1})\Phi(u\alpha + v) + u^{-1}(\phi(u\beta + v) \\
 &- \phi(u\alpha + v))
 \end{aligned}$$

with ϕ the standard Gaussian probability density function.

Proof. The result follows in a straightforward way by partial integration.

Remark. For particular choices of the parameters involved, (5) and (6) can be used to obtain explicit expressions for ES_k . In the next section two examples of this will be discussed.

4. NUMERICAL EXAMPLES

In this section we calculate ES_k for two combinations of parameter choices. Both cases correspond to situations that are of particular interest in practice.

4.1 Case 1

Use the following choices for the parameters

$k = 4$, that is bursts of 4 shots

$r = 1$ and $d = 0,5$

We take the rate of fire as 80 rounds per minute giving an interfiring time of $\epsilon = 0,75$ seconds.

In this case (1) becomes

$$\begin{aligned}
 (7) \quad c(t) &= 4ht - 2h \quad \text{for } 0,5 < t \leq 0,75 \\
 &= 4h - 4ht \quad 0,75 < t \leq 1,0
 \end{aligned}$$

Divide the unit interval into 4 disjoint intervals $(0; 0,25)$, $(0,25; 0,5)$, $(0,5; 0,75)$ and $(0,75; 1,0)$. Consider the time of fire of the first shot $T_1 = t$ to be

successively in each of these intervals. As t passes through these intervals we find the positions of T_2, T_3 and T_4 with respect to these intervals from (2). Denoting the intervals by J_1, J_2, J_3 and J_4 respectively, we have the following breakdown

$$(8) \quad \begin{aligned} t \in J_1 &\Rightarrow T_2 \in J_4 \Rightarrow T_3 \in J_3 \Rightarrow T_4 \in J_2 \\ t \in J_2 &\Rightarrow T_2 \in J_1 \Rightarrow T_3 \in J_4 \Rightarrow T_4 \in J_3 \\ t \in J_3 &\Rightarrow T_2 \in J_2 \Rightarrow T_3 \in J_1 \Rightarrow T_4 \in J_4 \\ t \in J_4 &\Rightarrow T_2 \in J_3 \Rightarrow T_3 \in J_2 \Rightarrow T_4 \in J_1. \end{aligned}$$

Clearly, from (8)

$$(9) \quad L(\{t : T_1 = t, T_j \notin (0, 5; 1, 0)\}) = 0, 5 \quad j = 1, 2, 3, 4.$$

Suppose we start with a positive peak. Note that for each T_j we have two intervals not containing peaks and two containing peaks (of which some are positive and some are negative). Denoting an interval without a peak by 0, an interval with a positive peak by + and a negative one by -, we can write (8) symbolically as (remembering that we take the first peak as positive),

$$(10) \quad \begin{array}{cccc} 0 & + & - & 0 \\ 0 & 0 & - & + \\ + & 0 & 0 & + \\ + & - & 0 & 0 \end{array}$$

Thus we have eight 0's, five +'s and three -'s. Note that starting with a negative peak will give the same but with the +'s and -'s interchanged.

Now, using (10) we see that in (3) we have four types of integrals

over J_3 with positive or negative peak

over J_4 with positive or negative peak.

Let us consider each of these in turn. Using the above lemma we have for the part with a positive peak.

$$(11) \quad \int_{J_3} \left[\Phi \left(\frac{b - c(t)}{\sigma} \right) - \Phi \left(\frac{-a - c(t)}{\sigma} \right) \right] dt$$

$$\begin{aligned}
&= \frac{1}{4} \left(1 - \frac{b}{h} \right) \Phi \left(\frac{b-h}{\sigma} \right) + \frac{b}{4h} \Phi \left(\frac{b}{\sigma} \right) \\
&\quad - \frac{1}{4} \left(1 + \frac{a}{h} \right) \Phi \left(\frac{-a-h}{\sigma} \right) + \frac{a}{4h} \Phi \left(\frac{-a}{\sigma} \right) \\
&\quad + \frac{\sigma}{4h} \left(\phi \left(\frac{b}{\sigma} \right) - \phi \left(\frac{b-h}{\sigma} \right) - \phi \left(\frac{a}{\sigma} \right) + \phi \left(\frac{a+h}{\sigma} \right) \right).
\end{aligned}$$

For the part with a negative peak we simply replace h by $-h$ in (11). Also, over J_4 we get from (7)

$$\begin{aligned}
(12) \quad &\int_{J_4} \left[\Phi \left(\frac{b-c(t)}{\sigma} \right) - \Phi \left(\frac{-a-c(t)}{\sigma} \right) \right] dt \\
&= \int_{J_4} \left[\Phi \left(\frac{b-4h+4ht}{\sigma} \right) - \Phi \left(\frac{-a-4h+4ht}{\sigma} \right) \right] dt \\
&= \int_{J_3} \left[\Phi \left(\frac{b-c(t)}{\sigma} \right) - \Phi \left(\frac{-a-c(t)}{\sigma} \right) \right] dt
\end{aligned}$$

by making the substitution $t \rightarrow \frac{3}{2} - t$. This means that for a positive peak the integrals over J_3 and J_4 are the same. Clearly this also holds for a negative peak.

Using the above results and adding terms we find that

$$\begin{aligned}
(13) \quad &ES_4(h, \sigma) \equiv ES_4 \\
&= 2 \left(\Phi \left(\frac{b}{\sigma} \right) - \Phi \left(\frac{-a}{\sigma} \right) \right) + 4 \int_{J_3} \left(\Phi \left(\frac{b-c(t)}{\sigma} \right) - \Phi \left(\frac{-a-c(t)}{\sigma} \right) \right) dt + 4 \int_{J_3} \left(\Phi \left(\frac{b+c(t)}{\sigma} \right) - \Phi \left(\frac{-a+c(t)}{\sigma} \right) \right) dt \\
&= 2 \left(\Phi \left(\frac{b}{\sigma} \right) - \Phi \left(\frac{-a}{\sigma} \right) \right) + h^{-1} \left\{ (b+h) \Phi \left(\frac{b+h}{\sigma} \right) + (a-h) \Phi \left(\frac{-a+h}{\sigma} \right) + \sigma \phi \left(\frac{b+h}{\sigma} \right) - \sigma \phi \left(\frac{-a+h}{\sigma} \right) \right\} \\
&\quad - \left\{ (b-h) \Phi \left(\frac{b-h}{\sigma} \right) + (a+h) \Phi \left(\frac{-a-h}{\sigma} \right) \right\}
\end{aligned}$$

$$+\sigma\phi\left(\frac{b-h}{\sigma}\right)-\sigma\phi\left(\frac{-a-h}{\sigma}\right)\Bigg\}].$$

Note in particular for the case of no peaks (that is letting $h \rightarrow 0$ in (13))

$$ES_4(0, \sigma) = 4 \left(\Phi\left(\frac{b}{\sigma}\right) - \Phi\left(\frac{-a}{\sigma}\right) \right).$$

In Table 4.1 we calculate ES_4 for given a, b, h, σ .

We consider two choices for range, namely 5 000 and 8 000 metres. For these ranges the values of σ were experimentally found to be 135 and 90 metres respectively. We also give results at two other choices of σ being roughly 10 % smaller and larger than the above. This gives some indication of the sensitivity of the results to the choice of σ . Two target sizes were used, corresponding roughly to a small and medium sized one. The effective target sizes (in metres) are (-27,117) and (-50,140) at range 5 000 metres and (-27,60) and (-50,83) at range 8 000 metres. Typical peaks of 5 and 10 minutes of arc were obtained in practice. These peak heights were transformed to metres using a ballistic trajectory model for the gunnery system, and are shown in Table 4.1. Here 5 and 10 minutes of arc correspond respectively to 120 and 240 metres.

For the results of Table 4.1 ES_4 was multiplied by 100 to give the expected number of hits in 100 bursts of 4 shots each.

These results are only for elevation. If we assume that there are no EORP's for training (as was the case in our application), then this adds a multiplicative constant to the above. However, since we are more interested in the relative change due to EORP's, this constant factor cancels out. The last column in Table 4.1 gives this percentage change compared to the case of no peaks.

From Table 4.1 we see that, especially for the 10 minute peaks, the EORP's can have quite a severe influence. There does not seem to be too much change over range. The effect due to changing σ is roughly between 6

and 9 percent with slightly higher percentages at the longer range and at smaller peak heights (for example changing σ from 120 to 135 metres at a range of 5 000 metres changes the number of hits from 170 to 155 giving a percentage change to $[(170 - 155)/170] \times 100 = 8,8\%$).

TABLE 4.1: CASE 1

Number of hits in 100 bursts of 4 shots each and percentage decrease in hit probability due to EORP's

<u>Range 5 000 m</u>		Peak Height (m)			Peak Height (m)	
		0	120	240	120	240
Standard deviation(m)	Effective target size(m)	Number of hits			Percentage decrease	
135	(-27, 117)	155	147	131	5,2	15,4
120	(-27, 117)	170	160	140	5,9	17,6
150	(-27, 117)	142	136	122	4,2	14,1
135	(-50, 140)	198	189	169	4,5	14,6
120	(-50, 140)	216	204	180	5,6	16,7
150	(-50, 140)	182	175	159	3,8	12,6
<u>Range 8 000 m</u>		0	75	150	75	150
90	(-27, 60)	146	139	125	4,8	14,4
80	(-27, 60)	162	153	135	5,6	16,7
100	(-27, 60)	133	128	116	3,8	12,8
90	(-50, 83)	213	204	184	4,2	13,6
80	(-50, 83)	234	222	197	5,1	15,8
100	(-50, 83)	195	188	172	3,6	11,8

4.2 Case 2

In this case we use the following choices for the parameters.

$k = 4$, that is bursts of 4 shots

$r = 1$ and $d = 1$,

that is continuous peaks. The rate of fire is again taken as 80 rounds per minute.

The calculations are much the same as in case 1. We again divide the unit interval into 4 sub-intervals. The time of fire of the first shot is taken successively in each interval and this leads, as in case 1, to the following symbolic representation

$$(14) \quad \begin{array}{cccc} + & + & - & + \\ + & - & - & + \\ + & - & + & + \\ + & - & + & - \end{array}$$

in the case of starting with a positive peak, and the + and - signs are interchanged when starting with a negative peak. By symmetry we see that the intervals over J_1 and J_4 and over J_2 and J_3 are equal. The function $c(t)$ is given by

$$(15) \quad \begin{aligned} c(t) &= ht & 0 < t \leq \frac{1}{2} \\ &= 2h - 2ht & \frac{1}{2} < t \leq 1 \end{aligned}$$

Using the above representation and the equality of the integrals, we find

$$(16) \quad \begin{aligned} ES_4(h, \sigma) &= \frac{1}{2} \left\{ 8 \int_0^{\frac{1}{2}} [\Phi((b - c(t))/\sigma) - \Phi((-a - c(t))/\sigma)] dt \right. \\ &\quad \left. + 8 \int_{\frac{1}{2}}^1 [\Phi((b + c(t))/\sigma) - \Phi((-a + c(t))/\sigma)] dt \right\}. \end{aligned}$$

Applying the lemma of Section 3 to (16), we obtain after some simplification that

$$(17) \quad \begin{aligned} ES_4(h, \sigma) &= 2\sigma h^{-1} \left\{ \frac{b+h}{\sigma} \Phi\left(\frac{b+h}{\sigma}\right) + \phi\left(\frac{b+h}{\sigma}\right) \right. \\ &\quad - \left(\frac{b-h}{\sigma}\right) \Phi\left(\frac{b-h}{\sigma}\right) - \phi\left(\frac{b-h}{\sigma}\right) - \left(\frac{a+h}{\sigma}\right) \Phi\left(\frac{-a+h}{\sigma}\right) \\ &\quad \left. + \phi\left(\frac{a+h}{\sigma}\right) - \left(\frac{h-a}{\sigma}\right) \Phi\left(\frac{h-a}{\sigma}\right) - \phi\left(\frac{h-a}{\sigma}\right) \right\}, \end{aligned}$$

which can be calculated easily for particular choices of a, b, h, σ . This is given in Table 4.2 for the same choices as in Table 4.1.

TABLE 4.2: CASE 2

Number of hits in 100 bursts of 4 shots each and percentage decrease in hit probability due to EORP's

<u>Range 5 000 m</u>		Peak Height (m)			Peak Height (m)	
		0	120	240	120	240
Standard deviation(m)	Effective target size(m)	Number of hits			Percentage decrease	
135	(-27, 117)	155	139	107	10,3	31,0
120	(-27, 117)	170	150	111	11,8	34,7
150	(-27, 117)	142	130	103	8,5	27,5
135	(-50, 140)	198	180	140	9,1	29,3
120	(-50, 140)	216	193	145	10,6	32,9
150	(-50, 140)	182	168	135	7,7	25,8
<u>Range 8 000 m</u>		0	75	150	75	150
90	(-27, 60)	146	132	103	9,6	29,5
80	(-27, 60)	162	144	107	11,1	34,0
100	(-27, 60)	133	122	99	8,3	25,6
90	(-50, 83)	213	195	154	8,5	27,7
80	(-50, 83)	234	210	160	10,3	31,6
100	(-50, 83)	195	181	148	7,2	24,1

In the last column we again give the relative change in the hit probability due to the EORP's.

As one would expect the decrease in hit probability in this case is even more severe than in case 1. The remarks following Table 4.1 also apply here except that the effect due to changing σ is roughly between 4 and 9 percent with the effect at a 10 minute peak being approximately 4 percent.

5. REMARKS

- Case 2 is clearly the more severe of the two. Comparing the last columns of Tables 4.1 and 4.2 we see that doubling the base of the peak (from 0,5 to 1,0) approximately doubles the percentage decrease in hit probability. The approximation holds very closely.
- The results calculated exactly in Section 4 can also be estimated by simulation. This was done for a few cases and the results were very similar to the above.
- In the model of Section 3 the ship movement was not included explicitly since this would have complicated matters unduly. As a simplified approximation to this we varied the peak height and peak duration within bounds that are realistic from real data.
- Only surface targets were considered since this is what our client was interested in. Air targets can be looked at in a similar fashion with the complicating factor that the standard deviation can (potentially) change from shot to shot as the range of the target changes. Also the component source of variation due to prediction becomes very important and severe biases can occur in certain scenarios.
- Spotting corrections were not included in the model but can be added fairly easily. One would expect this to aggravate the effect of the EORP's on hit probabilities.
- In this paper we assumed throughout that the range errors in fall of shot have a Gaussian distribution. This assumption is typically made and is a reasonable assumption for our situations.

- Not that we have employed a relatively simple model to describe the type of problem considered. It did, however, supply our client with information that was sufficient for him in his decision making process. More complex models can be considered for the problem but they do not necessarily add much to the quality of information provided.

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